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HHSKT: A learner–question interactions based heterogeneous graph neural network model for knowledge tracing☆

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ABSTRACT

Knowledge tracing (KT) has evolved into a crucial component of the online education system with the rapid development of online adaptive learning. A key component of the online education system, knowledge tracing (KT) assesses the state of knowledge by tracing each learner's learning activities. The deep KT model, however, is unable to completely extract the features of the questions and skills due to the heterogeneity of the knowledge structure and the sparsity of the interaction records. The model's capacity to handle diverse data is also restricted by over parameterization. Additionally, rather than focusing solely on a precise fit, Intelligent Tutoring System (ITS) should stress interpretable feedback to the learner. The deep KT approach's item parameters are still unable to give students useful feedback. This paper proposes to trace learner's short-term attentional knowledge based on heterogeneous hierarchical differentiation, named HHSKT. Hierarchical heterogeneous knowledge structures and short-term memory enhancement will be used to model the effects of different interaction sequences on learners. Specifically, knowledge structure features are extracted by constructing a heterogeneous graph-based graph information augmentation component. Question differentiation parameters are derived by transforming the TrueSkill system. Besides, learners' history-related practices are emphasized by windowing attention. Comparing regression-based and deep-based knowledge tracing experiments shows that HHSKT significantly outperforms the state-of-the-art approach on three real-world benchmark datasets (with an average AUC improvement of up to 3%), demonstrating the superiority of the proposed model.

1. Introduction

In recent years, online schooling on a massive scale has exploded, especially following the pandemic. Simultaneously, homogeneous online learning content cannot satisfy the individualized learning needs of a growing number of users (Injadat, Moubayed, Nassif, & Shami, 2020; Lan & Baraniuk, 2016). Current online learning platforms lack the adaptability to address these difficulties and provide learners with more specific, individualized feedback and help. The COVID-19-related school closures pushed schools to move instruction online and provide remote support for students (Boltz, Yadav, Dillman, & Robertson, 2021). Therefore, the traditional method of measuring the knowledge of learners' based on test scores and written responses lacks the necessary tools and environments to promote high-quality and thoughtful remote learning.

Data analysis and intelligent tutoring systems rely heavily on educational data mining (de Andrade, Rigo, & Barbosa, 2021). Knowledge Tracing (KT) is a learning situation analysis task based on computer-assisted large-scale data processing capabilities to trace learning activities, which can deliver diagnostic results without sophisticated human engagement for learners and teachers (Minn, Vie, Takeuchi, Kashima, & Zhu, 2021). Knowledge components (KCs) constitute learning content, and a learner's knowledge states (KS) are a collection of KCs. As demonstrated in Fig. 1, a learner's performance on the eleventh exercise can be estimated by tracing their mastery of KCs on the first ten exercises.

There are several approaches in the KT field that may be used to measure a learner's mastery of KCs, including bayesian methods (Zhao,

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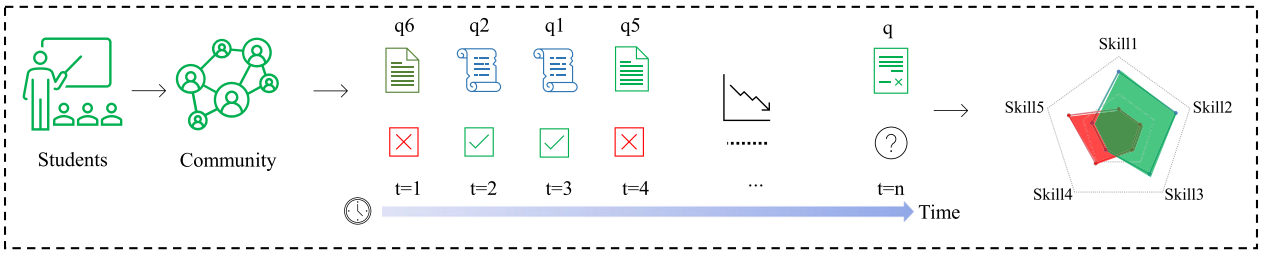


Fig. 1. An illustration of exercising process. The exercising process of a learner and the corresponding KS of each exercise.

Bhatt, Thille, Zimmaro and Gattani, 2020), logistic regression methods (Vie & Kashima, 2019), and deep learning methods (Piech et al., 2015). Psychometric approaches based on psychology (based on statistical regression, etc.) and Bayesian-based methods are typical examples of traditional methods (Ha, Hong, Hwang, & Yoon, 2018). Item response theory (IRT) is commonly employed in the KT framework (Su et al., 2021), where Rasch (1PL-IRT) is also employed to estimate item difficulty (Ghosh, Heffernan, & Lan, 2020). Some recurrent neural network (RNN) based methods have been proposed to model long dependencies between interactions (Ding & Larson, 2020; Lin & Chi, 2017; Piech et al., 2015), and transformer-based methods have been proposed to emphasize the weights of different KCs (Oya & Morishima, 2021; Pandey & Karypis, 2019; Pu, Yudelson, Ou, & Huang, 2020). These deep models have demonstrated impressive performance thanks to their robust fitting capabilities.

Deep knowledge tracing (DKT) is an early RNN-based deep KT model (Piech et al., 2015), while self-attentive knowledge tracing (SAKT) is the first model to integrate the transformer structure into KT (Pandey & Karypis, 2019). The RNN structure may step-by-step process sequential input from left to right and from bottom to top, retaining specific information of memory units depending on gating information and retaining the information recorded during steps t_0 to t_{n-1} to forecast t_n (Graves, rahman Mohamed, & Hinton, 2013). Online learning data consist mostly of sequential information, which is ideally suited for processing by RNN architectures. SAKT proposes an attention mechanism based on the sequential task in order to highlight the contribution of various KCs to the prediction of KC mastery at the present time. In addition to aforementioned models, graph neural networks (GNNs) have garnered significant attention (Veliković et al., 2017), and many graph construction methods have been proposed for models such as graph-based interaction knowledge tracing (GIKT) (Yang et al., 2021). Furthermore, the aggregation and representation capabilities of graph networks have been applied to significantly improve the prediction capability and interpretability of KT models substantially. Table 1 outlines the primary work and limits of the most prevalent KT techniques now in use, including some cutting-edge technologies.

Despite the fact that, current deep KT models are superior to traditional models for predicting learners' mastery of KC, the existing models have three major flaws. First, the relationships between items, skills, and learners in knowledge tracing datasets are often quite complex. Insufficient research exists on combining diverse features and large-scale learner interaction network architectures. Secondly, interpretable item parameters are frequently absent from deep model. It is more important for ITS to provide learners with practical and solvable feedback than model fitting capabilities. Lastly, DKT's modeling of learners' memory skills relies solely on gated information, which is insufficiently interpretable. The transformer-based approach can assign various weights to sequential characteristics. Attention has visual explanatory advantages, but nevertheless resists the memory process of the learner.

This work proposes HHSKT, a heterogeneous GNN-based model for multi-level summarization of knowledge structures and learner attention reinforcement. Higher-order relationships between items and skills are extracted by heterogeneous GNN, a rich representation of skill-level information is obtained by heterogeneous graph coarsening, and the final prediction is completed by item differentiation in conjunction with the attention mechanism of window sequence memory.

The following is a summary of our key contributions in this paper: (1) The model implements a higher-order item skill embedding representation of heterogeneous learner interaction graphs based on massive amounts of data. (2) The model transforms the TrueSkill system to obtain interpretable parameters. Explainable item differences and item difficulty are generated by generating learners with explainable feedback. (3) The model's windowed attention is superior to the mechanism's original implementation. Learners' short-term learning status can be appropriately highlighted by implementing a sequential positional attention method with windowed memory constraints.

The remainder of this paper is organized as follows. In Section 2, the work related to the model is briefly reviewed. In Section 3, the formulation of the model question is given and the overall structure of the model is described. In Section 4, the various modules in the model are explained in detail. In Section 5 the model is compared experimentally and discussed in detail and compared with the baseline model. The future research directions of this paper are expressed in Section 6.

2. Related work

This section provides a brief review of deep knowledge tracing, and related work consists of three main parts: the application of GNNs in the field of knowledge tracing, the TrueSkill system, and the study of learner memory mechanisms.

2.1. Knowledge tracing

Currently, the great majority of prevalent KT models are currently deep models, and this work focuses on deep knowledge tracing models as well. Knowledge tracing has traditionally focused on predicting the future performance of the learner and producing an evaluation of the learner's knowledge state (KS). Earlier knowledge tracing techniques, such as Bayesian Knowledge Tracing (BKT), depended on Hidden Markov Models (Zhao, Bhatt et al., 2020), Hidden Markov Models can forecast the probability distribution of hidden variables based on the present state of knowledge state (Hanghang, Xiaopeng, & Yuntao, 2021). In addition, learning factor analysis approaches (LFA) are essential knowledge tracing techniques (Yu, 2018; Zhang et al., 2021). There are several models in cognitive and psychometric factor analysis methodologies, such as IRT, additive factor model (AFM), performance factors analysis (PFA), etc. (Cen et al., 2006; Liao et al., 2012; Pavlik et al., 2009). Modeling and tracking learning behaviors by investigating factors related to learning capacities is referred to as factor analysis. The suggested knowledge tracing machine (KTM) framework combines IRT, AFM, and PFA models (Vie & Kashima, 2019). To predict the likelihood of learners' answers, KTMs utilize sparse sets of weights for each attribute. The classic LFA approach can may alter the investigated factors and so trace the knowledge states (KS) with a degree of flexibility, but is largely ineffective for long-interval tracing. The DASH and DAS3H bridge the gap between factor analysis models and memory models (Choffin et al., 2019; Lindsey et al., 2014). Knowledge tracing

Table 1
Existing major related work in the field of Knowledge Tracing.

Model	Interpretability	Advantage	Limitation
BKT (Zhao, Bhatt et al., 2020)	Self-Explaining	First use of Bayesian methods in the KT domain.	Insufficient personalization capability to be applied to large-scale data.
FBKT (Liu, Hu, Bu, & Yu, 2022)	Self-Explaining	Combine fuzzy states with BKT to trace continuous observations and fine-grained knowledge states.	It difficult to apply to large-scale online scenarios.
IRT (Su et al., 2021)	Self-Explaining	For the first time, psychometrics was combined with the KT field.	The training is difficult and the students' ability level is fixed.
MIRT (Chalmers, 2012)	Self-Explaining	Enriching the degree of feature involvement in IRT models.	The training is extremely difficult, and the feature dimensions are relatively fixed.
LFA (Yu, 2018)	Self-Explaining	LFA inherits and develops the Q matrix used to assess cognition in psychometrics and extends the theory of learning curve analysis.	The feature dimension is small and sensitive to the time factor.
AFM (Cen, Koedinger, & Junker, 2006)	Self-Explaining	Factor analysis methods were further developed to interpret the model predictions regarding statistical characteristics.	Fewer characteristic factors were used.
PFA (Pavlik, Cen, & Koedinger, 2009)	Self-Explaining		Fewer characteristic factors were used.
KTM (Vie & Kashima, 2019)	Self-Explaining	KTMs use a sparse set of weights for all features to model the learners' answer outcome.	The feature calculation is complex and has a cold start problem.
DASH (Lindsey, Shroyer, Pashler, & Mozer, 2014)	Self-Explaining	The DASH model is involved in memory forgetting modeling and factor analysis, where DASH stands for difficulty, ability, and student interaction history, respectively.	Inadequate response to multi-skilled problems.
DAS3H (Choffin, Popineau, Bourda, & Vie, 2019)	Self-Explaining	It combines aspects of IRT and PFA and extends the DASH model to calculate feature factors using a time-window-based counting function.	The window calculation is complex and has limited performance improvement.
Deep-IRT (Yeung, 2019)	Partially interpretable.	For the first time, item response theory (IRT) is combined with a knowledge tracing model based on a deep neural network structure of dynamic key–value memory network (DKVMN).	Due to the fixed binary Q-matrix designed by the expert, it lacks a personalized description of the student at the input level.
GKT (Nakagawa, Iwasawa, & Matsuo, 2019)	Black box model, graph structure can be interpreted based on local features.	First use of graph neural network methods in the KT domain.	Lack of ability to deal with multiple concepts and complex relationships between conceptual problems.
GIKT (Yang et al., 2020)		The problem of data sparsity is solved by using GCN to aggregate embedding representations under concept and exercise relations.	The model is complex, the training scale is large, and it is not pedagogically interpretable.
PEBG (Liu et al., 2020)		Graph-based pre-trained skill features are used to enrich the feature information used by the model.	The features of the graph structure are not well taken into account.
HGKT (Tong, Wang, Liu, Zhou and Han, 2020)	Black box models, interpreting models by interpreting structured features.	Hierarchical exercise graphs are constructed based on hierarchical relationships between exercises (direct and indirect relationships).	Knowledge concepts are divided into different granularities and are not sufficiently hierarchical.
JKT (Song et al., 2021)		The connection of exercises across concepts is established to capture high-level semantic information while increasing the interpretability of the model.	Insufficient consideration of the directionality of prerequisites between knowledge and the correlation information between nodes.
SKT (Tong, Liu et al., 2020)		A propagation model is introduced to model the propagation of different knowledge point cascades of influence.	
SAKT (Pandey & Karypis, 2019)	Black box model with local interpretation based on attention.	For the first time in the KT domain, the transformer model is used to model student interaction history based on a self-attentive mechanism.	Other features around the attention of the nodes are suppressed and the order between exercises is not taken into account.
SAINT (Shin et al., 2021)		Two multiple attention layers were overlaid further to capture the complex relationship between practice and response.	
AKT (Ghosh et al., 2020)		AKT perceives the context between questioning and skills through attentional mechanisms.	

based on ensemble learning is another essential KT subdiscipline (Zia et al., 2021). However, ensemble learning is a stable learning technique and the variety of individual students must be increased (Jiang et al., 2021; Jiang, Yu, Gong, & Du, 2022). In addition, deep models provide excellent interpretability when integrated with factor analysis models, such as, Deep-IRT and NeuralCD, which combine IRT with deep approaches (Wang et al., 2020; Yeung, 2019).

The majority of data in the area of education are non-Euclidean structured graph data, and numerous approaches have been developed to handle graph-structured data (Veliković et al., 2017; Zhao, Song et al., 2020). Graph convolutional networks (GCN) are inspired by CNN ideas to update their own representation based on neighboring nodes on graph-structured input, which may be aggregated to higher-order neighbor information via multi-layer convolution (Schlichtkrull et al., 2018). GNN has been implemented in several disciplines, including natural language processing (NLP), recommendation systems, and others. Ahale, Pattanshetti, and Nayak (2021) and Schlichtkrull, Cao, and Titov (2021). There are GNN applications in the KT domain, such as graph knowledge tracing (GKT) (Nakagawa et al., 2019), but they do not take learner characteristics into account. A heterogeneous graph neural network is used in HHSKT to extract higher-order representations of model inputs and to ensure the performance of the model on large-scale data through subgraph sampling. Hierarchical features are the information underlying the learner's educational process, and hierarchical information is crucial to the educational process (Shuai, Li, Liu, & Shen, 2022). Clustering in hierarchical graph KT (HGKT) (Tong, Wang et al., 2020) introduces hierarchical aspects of items, although hierarchical features produced by threshold clustering do not adequately capture hierarchical features of items. One technique utilizes undulating edge matching heuristic, while the other coarsening method picks a subset of nodes from the original graph to create a coarse graph node set (Ma & Chen, 2021). A heterogeneous graph neural network is used in HHSKT to extract higher-order representations of model inputs and to ensure the performance of the model on large-scale data through subgraph sampling.

2.2. TrueSkill system

The TrueSkill rating system is primarily designed to evaluate gamers as a derivation of the Glicko scoring system (Herbrich, Minka, & Graepel, 2007; Minka, Cleven, & Zaykov, 2018; Quispe & Luna, 2015), but its excellent rating capabilities are equally applicable to measuring learning capacity in non-gaming situations (Lee, 2019). The TrueSkill methodology is superior to the PageRank method for predicting question difficulty in community Q&A (Wang, Liu, Wang, & Guo, 2014), such as stackoverflow's question difficulty classification. In a large-scale speaker-likability rating test, Baumann showed the reliability of TrueSkill method utilizing score rankings (Baumann, 2017). In social scenarios, students interacted with graph-based scenarios to develop practical interpersonal skills in unexpected contexts. TrueSkill can reliably predict the decisions made by students in simulated social situations (Kawatsu, Hubal, & Marinier, 2018). The preceding study illustrates the consistency and precision of the TrueSkill method. The origins of item response theory lie in the field of psychometrics. It posits that there is an underlying trait that reflects each learner's ability and that this attribute may be detected through the learner's item responses. Item response theory (IRT) consists of two factors, ability θ and difficulty b_i, with the higher item's difficulty matching to the higher the learner's necessary ability (Harvey & Hammer, 1999). By clearly defining factors such as item difficulty and student competence, the model's computational process is visible and easily interpretable. However, IRT assumes that students' skills remain constant over time, which is inappropriate for long-term knowledge tracing. Rasch model may be considered as a 1PL-IRT, however when the number of learners is relatively small (1, 2.5, and 5 percent of the total batch), IRT cannot evaluate learners' abilities or forecast their performance in question-solving. In addition, the area under curve (AUC) values from the IRT model declined gradually until 50% of the data were utilized (Lee, 2019). Therefore, the TrueSkill system is incorporated into the depth model to correctly analyze input data in tiny batches.

2.3. Mechanisms of attention and forgetting

It has been demonstrated that attention processes are useful for sequential activities (Vaswani et al., 2017). SAKT integrates the attention mechanism with knowledge tracing to enhance the accuracy of model prediction via attention interpretation. Dynamic key–value memory networks (DKVMN) is more explicative and provides a knowledge concept matrix and a knowledge state matrix to dynamically track the mastery of knowledge concepts by learners (Zhang, Shi, King, & Yeung, 2017). The SAINT model creates a multilayer attention mechanism and identifies a more appropriate technique for generating Q, K, and V values for KT (Shin et al., 2021). The exercise aware KT (EKT) model is an extension of the exercise-enhanced recurrent neural network (EERNN) that incorporates an LSTM-based attention mechanism and models the link between the underlying knowledge concepts to increase the accuracy of model predictions (Liu et al., 2021). Transformer is also utilized for the RKT model (Pandey & Srivastava, 2020). Nevertheless, provide a relational-aware self-attentive layer that blends textual content of practice connection information, learner performance data, and contextual information on forgetting behavior described by an exponentially decaying kernel function. With the above work, it was demonstrated that different attention weights may accurately represent the memory characteristics of learners. On the other hand, Murre mathematically shown that if the distribution of learning rates follows a gamma distribution, a uniform distribution, or a half-normal function, the power function behaves as the average result of an exponential function (Murre & Chessa, 2011). This means that the general forgetting pattern of the learning process may be seen as following a power function distribution when learning knowledge points on a large scale, and HawkesKT demonstrated the importance of temporal crossover effects of memory in KT tasks (Wang et al., 2021). The present attention mechanism has demonstrated efficacy in the KT domain. However, existing attentional processes do not precisely correspond to the memory characteristics of learners, and the contribution of distant interaction records to the present moment should be reevaluated. The attention mechanism can efficiently extract focused information but concurrently introduces invalid distant information simultaneously. A windowed attention mechanism is proposed to capture the learners' short-term characteristics of learners, and it is demonstrated that this mechanism can effectively improve the model's AUC performance.

3. Problem definition

The KT model takes the sequence of learner–question interactions as input, and by looking at the interactions that came before the current one, it predicts the learner's response to the following question. As a result, it can be considered as an autoregressive problem or a time-series prediction problem. The knowledge tracing process is defined as follows. The notations are listed in Table 2 for the sake of presentation clarity.

The learner's learning record mainly includes information such as questions, answers, correctness, and response time. The learning state of the learner s can be represented as a triad of questions, concepts and answers (q_i^s, c_i^s, r_i^s). So the learning sequence can be represented as $\{(q_1, c_1, r_1), (q_2, c_2, r_2)...(q_i, c_i, r_i)\}$, where $q_i^s \in \mathbb{N}^+$ is the question number encountered in the learning process, $c_i^s \in \mathbb{N}^+$ represents the knowledge concept corresponding to the question, and $r_i^s \in \{0, 1\}$ indicates whether the learner has answered the question correctly. The input sequence is constructed by sampling as a graph $BG = (V, E)$. Each question q_i corresponding to N knowledge concepts is constructed

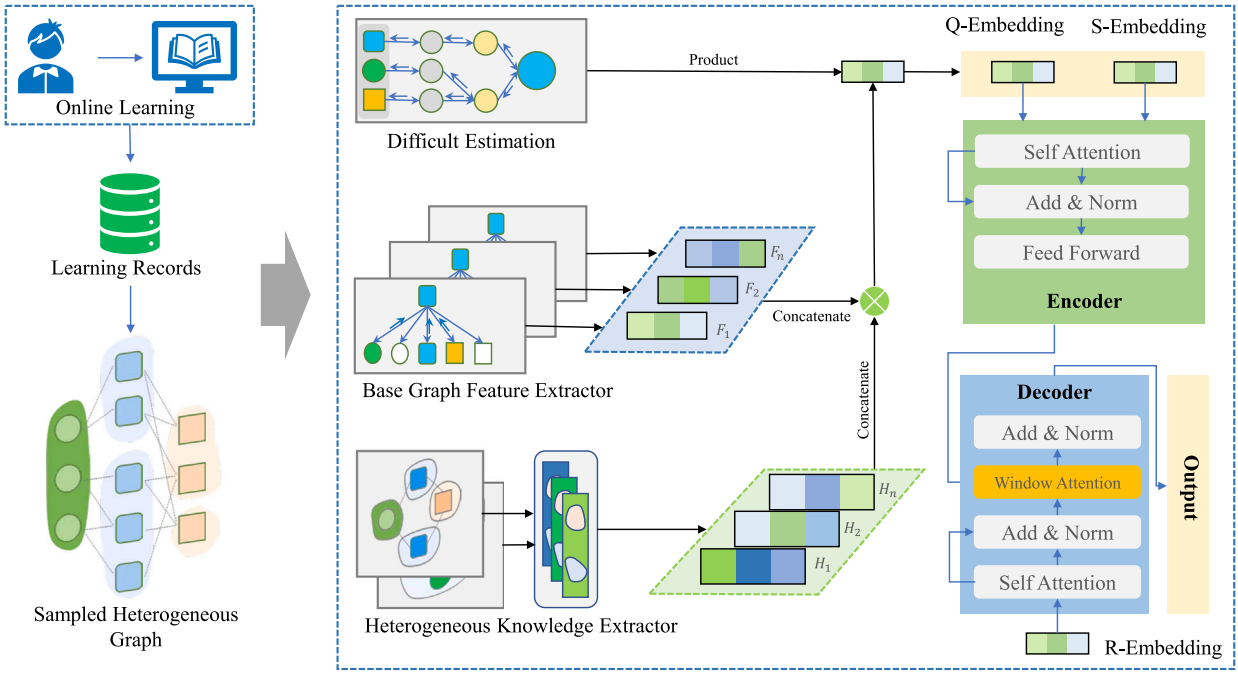


Fig. 2. The middle part is the base feature extractor (BFE), and the bottom part is the hierarchical feature extraction module (HFE). The sequence processing module is the main module for extracting sequence features, which ultimately predicts learners' performance based on their learning behavior and the item difficulty evaluator (DE) combined with the representation information from the BFE and the HFE.

Table 2
Table of notations.

Notations	Descriptions
KT	Knowledge Tracing;
KS	Knowledge State;
KCs	Knowledge Components;
GNN	Graph Neural Network;
GCN	Graph Convolutional Network;
RNN	Recurrent Neural Network;
BG	Basic graph;
BFE,HFE	Base Feature Extractor; Hierarchical Feature Extractor;
q_i^s, c_i^s, r_i^s	Representing the questions, concepts and responses answered by the learner s at time step i, respectively;
V	An adjacency matrix containing three types of nodes: question nodes, knowledge nodes, and learner nodes;
E	The connections of nodes contained by V;
λ	Node features and Edge features;
x_i	The original input sequence;
u_i, q_i, s_i	Learner u_i; Question q_i; Skill s_i;
d_i	The difficulty evaluator output at time step i;
att_i	The window attention at time step i;
H_t^l	The node representation of node t at the lth GNN layer;
$ATT-head^i$	Heterogeneous multi-head attention in BFE;
$Msg-head^i$	Heterogeneous multi-head message in BFE;
$W_{\phi(e)}^{ATT}$	Edge-based weight matrix that capture different relational information;
$\hat{A}$, X, W_α	Adj matrix, Node embedding, Weight matrix;
S, Z	The coarsening matrix; The intermediate hidden vector of the graph coarsening process;
μ, σ	Normal distribution parameters: the mean μ and the variance σ;

as a node $V = \{v_1, \ldots, v_N\}$, which share dependencies called edges $E \in \{V \times V\}$, and the same skill s_i usually has connected edges with multiple questions, and the same learner u_i is associated with multiple questions. In addition, different edges E_i usually occur at different time nodes, so the lag time (LT), and answer time (RT) in the interaction sequence should be considered as properties of the edges. The input sequence $\mathcal{X} = \{x_1, x_2, \ldots, x_t\}$ are sampled and aggregated with higher-order information by graph BG to form the embedding $\hat{X} = \{\hat{x}_1, \hat{x}_2, \ldots, \hat{x}_t\}$, and the final prediction $r_i^s \in \{0, 1\}$ is obtained by the prediction module, which described in Section 4.

4. The HHSKT model

4.1. Model overview

The architecture of our proposed method is shown in Fig. 2. HHSKT consists of two main parts: the question characterization system and

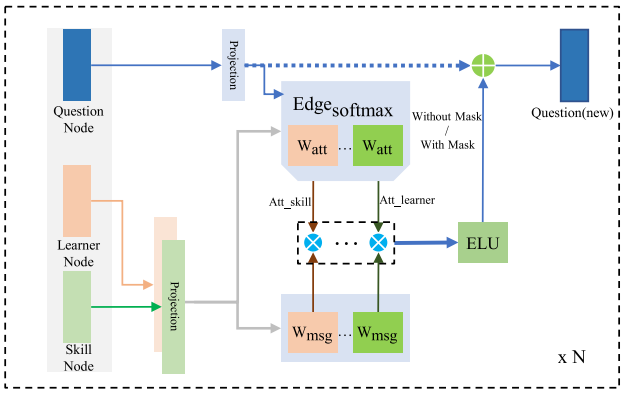


Fig. 3. The GNN-based base feature extractor (BFE) is used to extract the interaction features and knowledge structure features in the heterogeneous graph. The update process of the question node embedding is shown in the figure, where the blue part and non-shaded symbols indicate the target and neighboring nodes.

the sequence processing system. The details of these processes are explained below.

Here, the question characterization consists of two modules, the bottom one is the base feature extractor (BFE), and the other is the hierarchical feature extraction module (HFE). The sequence processing module is the main module for extracting sequence features, which ultimately predicts learners' performance based on their learning behavior and the item difficulty evaluator (DE) combined with the representation information from the BFE and the HFE. Moreover, the essential step of our model is to generate the basic graph (BG) from the exercise, where the basic graph is the heterogeneous graph $BG = (V, E, \lambda)$. V is an adjacency matrix containing three types of nodes: question nodes, knowledge nodes, and learner nodes, the connections of these nodes are saved as E, and λ represents node features and edge features.

The original input sequence $\mathcal{X} = \{x_1, x_2, \ldots, x_t\}$, each x_i including learner u_i, question q_i and skill s_i, the input sequence is first transformed from each input token x_i into an embedding $\hat{x}_i$ using an embedding layer (omitted in the figure), where $\mathbf{E}_s \in \mathbb{R}^{|\mathcal{S}|\times d}, \mathbf{E}_q \in \mathbb{R}^{|\mathcal{Q}|\times d}, \mathbf{E}_a \in \mathbb{R}^{2\times d}$. The original input sequence is also sampled to form a Sampled Heterogeneous Graph, after which a higher-order representation of the input sequence is obtained by BFE aggregation $\hat{X} = \{\hat{x}_1, \hat{x}_2, \ldots, \hat{x}_t\}$, but not directly using the graph hidden states for prediction. An additional transformer layer is used to fuse the difficulty of the difficulty evaluator output $\hat{D} = \{\hat{d}_1, \hat{d}_2, \ldots, \hat{d}_t\}$ and the window attention $\hat{ATT} = \{\hat{att}_1, \hat{att}_2, \ldots, \hat{att}_t\}$, the predicted response to the next question q_t is computed from the information in the first $t-1$ elements. The model forms a prediction tensor $\mathcal{B}\times\mathcal{H}\times\mathcal{W}$ through a fully connected layer, where each element represents the probability that the learner has mastered the corresponding knowledge concept at time t.

4.2. Base feature extractor

Formally, given an input graph $BG = (V, E, \lambda)$ and a GNN model GNN_θ, the key idea of majority graph neural networks (GNNs) is to aggregate feature information from a node's direct (first-order) neighbors, such as inductive representation learning on large graphs (GraphSAGE) or Graph Attention network (GAT) (Hamilton, Ying, & Leskovec, 2017; Veliković et al., 2017). However, they cannot aggregate information from different nodes and are vulnerable to neighbor nodes (Zhang, Song, Huang, Swami, & Chawla, 2019). A basic feature extraction module based on a heterogeneous GNN approach is used for question feature and skill processing, as shown in Fig. 3. The extractor takes the item sequence $\{q_1, q_2, \ldots, q_i\}$ and skill sequence $\{c_1, c_2, \ldots, c_i\}$ as inputs. Heterogeneous graphical information is used as additional input, learner nodes and interactions between learners, items and responses are included. The output of the feature extractor module is an embedding of items and skills, containing the higher-order features after being crossed.

The first contribution based on the intuition is that the differentiation of skills and items is not only related to themselves, but also to the interaction sequence. The original graph is heterogeneous, and more information about the features related to the current input sequence can be found on the graph from the input items and skills. So inspired by Heterogeneous Graph Transformer (HGT), the global sampling was introduced in base feature extractor (BFE) which based on multi-hop sampling of small batches of input sequences, to obtain feature data according to richness (Hu, Dong, Wang, & Sun, 2020). After multi-hop sampling, the sampled data and the original data are fed separately into a heterogeneous graph neural network for information aggregation, where different skills and item embeddings contain features of their neighboring related nodes. The raw data use masks to prevent the relationships to be predicted from being captured in advance.

4.2.1. Heterogeneous knowledge extractor

The basic extractor is trained and sampled to obtain the relative difficulty information between different questions, as shown in Fig. 3.

Formally, the original data needs to be projected through the embedding layer into the desired embedding information. In this module, the embedding layer is represented as $\mathbf{E}_s \in \mathbb{R}^{|\mathcal{S}|\times d}, \mathbf{E}_q \in \mathbb{R}^{|\mathcal{Q}|\times d}, \mathbf{E}_a \in \mathbb{R}^{2\times d}$. $\mathcal{S}$ and $\mathcal{Q}$ represent the number of skills and items, respectively. The module suppose H_t^l is the node representation of target node t at the lth GNN layer, the update procedure from the $(l-1)$th layer to the lth layer shown as Eq. (1):

$$H_t^l \leftarrow \underset{\forall s\in N(t), \forall e\in E(s,t)}{\text{Aggregate}} \left(\left\{ \text{Extract}\left(H_s^{(l-1)}; H_t^{(l-1)}, e \right) \right\} \right) \tag{1}$$

where $N(t)$ denotes all the source nodes of node t and $E(s,t)$ denotes all the edges from node s to t. $Extract(\cdot)$ represents the neighbor information extractor. For example, other questions, skills, and learners are all possible source nodes for it when question is the target node. Node attention uses the target node's representation H_t^{l-1} and the edge e between the two nodes as query, and extract useful information from source node H_s^{l-1}. $Aggregate(\cdot)$ serves as the aggregation function of the neighborhood information.

The messages are calculated using different messages for different edge weights inspired by HGT and Transformer. The weight matrices of the different interaction operators are parameterized as source node projection, edge projection and target node projection.

$$\text{Attention}(s,e,t) = \underset{\forall s\in N(t)}{\text{Softmax}} \left(\|_{i\in[1,h]} \text{ATT}-head^i(s,e,t) \right) \tag{2}$$

$$\begin{aligned} \text{ATT-head}^{\,i}(s,e,t) &= \left(K^i(s) W_{\phi(e)}^{ATT} Q^i(t)^T \right) \cdot \frac{\omega_{\langle\tau(s),\phi(e),\tau(t)\rangle}}{\sqrt{d}} \\ K^i(s) &= \text{K-Linear}^{\,i}_{\tau(s)} \left(H^{(l-1)}[s] \right) \\ Q^i(t) &= \text{Q-Linear}^{\,i}_{\tau(t)} \left(H^{(l-1)}[t] \right) \end{aligned} \tag{3}$$

Different attention vectors are computed for different edges, and a multi-headed attention mechanism is used, as shown in Eqs. (2) and (3). For the ith attention head $ATThead_i(s,e,t)$, The source node s is projected into the ith key vector $K^i(s)$ using different linear depending on the type of the source node. Each type of node in the graph has a corresponding unique Linear projection, thus keeping the differences between different types of nodes to the maximum extent. For embedding of items, the mask matrix is introduced to prevent the information of edges from being aggregated into the node representation in advance. The computation of dot product attention between different nodes for heterogeneous graphs is different from the original transformer model. $\omega \in \mathbb{R}^{|\mathcal{A}|\times|\mathcal{R}|\times|\mathcal{A}|}$ represents the weights of different node relationships, and links from different relationships usually have different weights for the target node. There may be different edge types, such as $\tau(s)$ and $\tau(i)$, between items and skill nodes, and between learners and items. Therefore, each edge type $\phi(e)$ retains a unique edge-based matrix $W_{\phi(e)}^{ATT} \in \mathbb{R}^{\frac{d}{h}\times\frac{d}{h}}$ to capture different relational information.

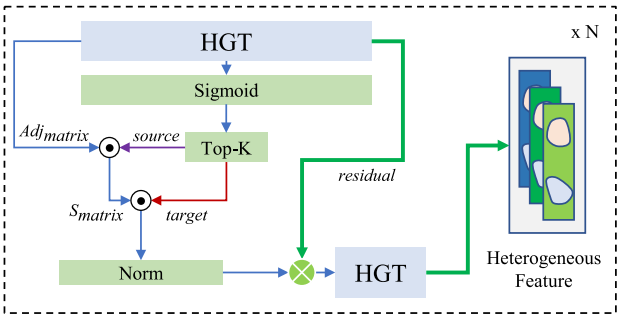


Fig. 4. Hierarchical feature extractors (HFE) are formed by heterogeneous graph node coarsening operations. The hierarchical feature extractor is formed by a heterogeneous graph node coarsening operation. Node features are extracted by HGT operator, after that nodes are selected and ranked according to the importance of major node features. Besides, the heterogeneous graph structure features are obtained by connecting different hierarchical features through residuals and normalizing them.

4.2.2. Interaction message passing

Message passing is the key operation of GNN, which aims to pass information from the source node to the target node. The meta-relationships of edges are added during message passing to mitigate the differences in the distribution of different types of nodes and edges. Messaging of learner interaction data is presented in Eqs. (4) and (5).

$$\text{Message}_{(s,e,t)} = \|_{i\in[1,h]} Msg\text{-head}^{i}(s,e,t) \tag{4}$$

$$\text{Msg-head}^{i}(s,e,t) = \text{Linear}^{i}_{\tau(s)}\left(H^{(l-1)}[s]\right) W^{MSG}_{\phi(e)} \tag{5}$$

The information of the ith message head $MSGhead_i(s,e,t)$ comes from the rest of the nodes linked by the target node, and the $W^{MSG}_{\phi(e)} \in \mathbb{R}^{\frac{d}{h}\times\frac{d}{h}}$ is used to measure the edge weights, where h means the count of multi head. The $Message(s,e,t)$ corresponding to each node is obtained by merging all message heads.

4.2.3. Message aggregation

After completing the computation of attention and messages, the new messages need to be aggregated to the target node to complete the whole update process. The new node features are formed by joining the residuals of the original features with the new features, as Eqs. (6) and (7).

$$\widetilde{H}^{(l)}[t] = \bigoplus_{\forall s\in N(t)} (Attention(s,e,t)\cdot Message(s,e,t)) \tag{6}$$

$$H^{(l)}[t] = Linear_{\tau(t)}\left(\sigma\left(\widetilde{H}^{(l)}[t]\right)\right) + H^{(l-1)}[t] \tag{7}$$

Obtain the hidden representation $H^{(l)}[t]$ of the target t after multiplying the attention and the message. The new node features contain all attention weights and messages, and the feature information of the target node is updated through residual connections.

4.3. Hierarchical feature extractor

Our second intuition is that the skill structure information provided in the dataset does not fully represent the true level of skill differentiation. A hierarchical feature extractor is used to obtain structured and hierarchical input features, as shown in Fig. 4.

4.3.1. Heterogeneous coarsening

In the HGKT model, hierarchical information has been shown to be crucial for knowledge tracing text (Tong, Wang et al., 2020). However, the hierarchical information obtained by clustering does not fully reflect the real question hierarchical information. Accurate hierarchical features are obtained by heterogeneous graph coarsening using a heterogeneous graph as input. Specifically, the coarsening information of different types of nodes is extended to heterogeneous graphs by computing them to obtain a rich representation of hierarchical features.

Between learners and skills are related not only through node information, but also through graph structure information. To utilize node features and subgraph features, graph convolution is used to compute a vector $\alpha \in \mathbb{R}^{n\times 1}$ that weights all nodes. Eqs. (8) and (9) demonstrates the node-based coarsening process.

$$\alpha = \text{sigmoid}\left(\widehat{A}XW_{\alpha}\right) \tag{8}$$

$$S = \ell_1 - Norm\left[\widehat{A}_s \odot \alpha_s^T\right] \tag{9}$$

$X \in \mathbb{R}^{n\times d}$ is the node feature matrix and $W_{\alpha} \in \mathbb{R}^{d\times 1}$ is the weight vector. In order to coarsen to m nodes, the first m values of α are selected and listed in sorted order. Denote such a vector by $\alpha_s \in \mathbb{R}^{m\times 1}$, where the subscript s denotes sorted. $\widehat{A}_s \in \mathbb{R}^{n\times m}$ denotes the selected and sorted version of $\widehat{A}$. $\ell_1 - Norm$ indicates that S is obtained by row-wise normalization. The final coarsening skill node feature representation is shown in Eq. (10).

$$Z = \text{GNN}(A,X), Z_c = S^T Z, X_c = \text{GNN}\left(A_c, Z_c\right) \tag{10}$$

where S denotes the coarsening matrix. Z denotes the intermediate hidden vector of the graph coarsening process. X_c denotes the final graph signal obtained after graph coarsening.

4.4. Sequence feature modeling

The processing of serialized data by the model is based on the Transformer model. Since the original dot product attention is not consistent with human short-term memory ability and cannot be fully applied to the learner ability retrospectively, HHSKT has improved the calculation of attention by windowing.

4.4.1. Window attention mechanism

The application of the HawkesKT model to temporal effects suggests that short-term memory repeated stimuli are very helpful in predicting learners' responses (Howard & Kahana, 2002; Mei & Eisner, 2017; Wang et al., 2021), and the importance of short-term memory is demonstrated in the AKT model (Ghosh et al., 2020). However, a series of questions is learned by learners in the learning system, and the serial position effect suggests that learners are likely to accept only the beginning and the end of the content. Therefore, HHSKT improves prediction performance by emphasizing the importance of the first and last positions of a particular question. As shown in Eqs. (11) and (12).

$$\alpha_{t,\tau} = \frac{\exp\left(s_{t,\tau}\right)}{\sum_{\tau'}\exp\left(s_{t,\tau'}\right)} \tag{11}$$

$$s_{t,\tau} = \frac{\left(e^{-\gamma\cdot d(t,\tau)} - \gamma * \frac{e^{d(t,\tau)}-e^{-d(t,\tau)}}{e^{d(t,\tau)}+e^{-d(t,\tau)}}\right)\cdot \mathbf{q}_t^{\top}\mathbf{k}_{\tau}}{\sqrt{D_k}} \tag{12}$$

where γ is the learnable decay rate parameter and $d(t,\tau)$ is the time distance metric between time steps t and τ.

4.4.2. Difficulty estimation

The difficulty of the problem belongs to the original nature of the question and needs to be taken into account. The Rasch model (Holster & Lake, 2016) is generally regarded as a 1PL-IRT model, and it has been proven effective for question embedding (Ghosh et al., 2020). The TrueSkill system is a scoring system based on Bayesian inference (Herbrich et al., 2007). TrueSkill system assumes that a player's level can be represented by a normal distribution, which can be described by two parameters: the mean μ and the variance σ. Lee's work demonstrates that the TrueSkill system can correctly estimate question difficulty features (Lee, 2019). Learners, items and skills are considered as different game participants and a response of 1 identifies the learner

as the winner and vice versa, and the update process for items and skills is shown as (13).

$$\begin{aligned}
\hat{\mu}_{\text{winner}} &\leftarrow \hat{\mu}_{\text{winner}} + \frac{\hat{\sigma}^2_{\text{winner}}}{c} \cdot v\left(\frac{\hat{\mu}_{\text{winner}} - \hat{\mu}_{\text{loser}}}{c}\right) \\
\hat{\sigma}^2_{\text{winner}} &\leftarrow \hat{\sigma}^2_{\text{winner}} \sqrt{1 - \frac{\hat{\sigma}^2_{\text{winner}}}{c^2} \cdot w\left(\frac{\hat{\mu}_{\text{winner}} - \hat{\mu}_{\text{loser}}}{c}\right)} \\
\hat{\mu}_{\text{loser}} &\leftarrow \hat{\mu}_{\text{loser}} - \frac{\hat{\sigma}^2_{\text{loser}}}{c} \cdot v\left(\frac{\hat{\mu}_{\text{winner}} - \hat{\mu}_{\text{loser}}}{c}\right) \\
\hat{\sigma}^2_{\text{loser}} &\leftarrow \hat{\sigma}^2_{\text{loser}} \sqrt{1 - \frac{\hat{\sigma}^2_{\text{loser}}}{c^2} \cdot w\left(\frac{\hat{\mu}_{\text{winner}} - \hat{\mu}_{\text{loser}}}{c}\right)}
\end{aligned} \tag{13}$$

v and w represent the mean additive truncated gaussian function and the variance multiplicative function, respectively. c is a normalized factor, denoted as $c^2 = 2\beta^2 + \sigma^2_{\text{winner}} + \sigma^2_{\text{loser}}$. The TrueSkill scores are continuously updated during training using small batches of data, and this parameter is not updated during prediction. As shown in Eq. (14).

$$\begin{aligned}
d_\tau &= \text{Softmax}\left(Linear_{\tau(t)}\left([\mu_\tau, \sigma_\tau]\right)\right) \\
inputs_{x_t} &= [E_{s_t} \cdot \mathbf{d}_{s_t}, E_{q_t} \cdot \mathbf{d}_{q_t}]
\end{aligned} \tag{14}$$

After completing the scoring, μ and σ are mapped into one-dimensional vectors by linear projection, and then the final difficulty prediction features are obtained by softmax.

4.4.3. Model training

To evaluate the difference between the final prediction of the model and the dataset labels, the negative log likelihood of the observed series was used as the loss function.

$$\mathcal{L} = -\sum_{i \in I}\left(r_i \log\left(p_i\right) + \left(1 - r_i\right)\log\left(1 - p_i\right)\right) \tag{15}$$

Specifically, at each epoch of the training process, the training loss of the model is updated by minimizing the cross-loss entropy of the predicted value p_i and the true value r_i, and AdamW is used as an optimization function in the model.

5. Experiments

In this section, three open source datasets are used to conduct extensive experiments to demonstrate the effectiveness of the HHSKT model in terms of performance metrics and visualization.

5.1. Implementation details

For accurate evaluation of the model, 5-fold cross-validation was used in the experiments, where 20% within each fold was used for testing, 20% for validation, and 60% for training. Experiments were set to stop early to prevent overfitting of the model. The original code and hyperparameters were set for all baselines. In this experiment, the area under the ROC curve (AUC) and the accuracy of prediction (ACC) are used as the evaluation metrics, and the larger the AUC and ACC, the better the prediction performance of the model.

Since the length of learners' time series varies significantly among different data sets, the response series are uniformly intercepted to a length of no more than 200 for the efficiency of the model. Parts longer than the specified length will be truncated to rearrange into a new set of data. The learning rate of our model is set to 0.001, and the batch size for training datasets is set to 128. We use AdamW to optimize our training process. The limitation of the early stopping module is ten epochs. All tests are carried out on the NVIDIA Tesla V100S, which serves as the model's training platform.

Table 3
Datasets.

Dataset	Student	Question	Skill	Interaction
ASSIST2009	3.7k	16k	123	260k
ASSIST2017	7k	3k	101	942k
Junyi15	24k	835	41	25,797k

5.1.1. Datasets

We use three open-source datasets and seven KT models as baselines for our experiments, which are described below.

- **ASSISTments2009** The ASSISTments2009 dataset is an open source dataset provided by the ASSISTment platform and is also the benchmark dataset for most knowledge tracing models (Feng, Heffernan, & Koedinger, 2009). In our experiment, this dataset contains 258,149 records and 123 KCs.
- **Junyi15** This dataset comes from Junyi Academy, an E-Learning platform based on Khan Academy (Chang, Hsu, & Chen, 2015). There are 247,605 learners, 835 questions and 41 skills.
- **ASSISTments2017** This competition uses data from a longitudinal study, now over a decade long, led by Professor Ryan Baker and Professor Neil Heffernan (Patikorn, Baker, & Heffernan, 2020). We use 942k records of 7k learners and 3k questions with 101 skills in our experiment.

Records of learners with less than 5 interactions were excluded from the data preprocessing. The statistical information of all datasets used in this paper is shown in the Table 3.

5.1.2. Baselines

The model is compared with several models to demonstrate the advancedness of the model proposed in this paper.

- **IRT:** IRT (Item Response Theory) (Su et al., 2021) is a paradigm for the design, analysis, and scoring of tests, questionnaires, and similar instruments measuring abilities, attitudes, or other variables. It is considered to be effective in estimating the difficulty of the question.
- **PFA:** PFA is formed by extending the AFM model, which divides learners' interaction in the learning process into positive and negative aspects. The AFM model can be seen as a particular case of the PFA model (Cen et al., 2006; Pavlik et al., 2009).
- **DAS3H:** DAS3H builds on the AFM and includes a representation of the temporal distribution of past practice for the skills covered by the project. Specifically, DAS3H allows the learning and forgetting curve to vary from one skill to another (Choffin et al., 2019; Lindsey et al., 2014).
- **DKT:** DKT is the first application of deep recurrent neural networks in knowledge tracing. We use the LSTM implementation of this model as a benchmark (Piech et al., 2015).
- **SAINT+:** SAINT+ is based on the Transformer model, incorporating interaction time features and lag-time features to model short-term memory and forgetting behavior. It is a further developed version of SAINT (Shin et al., 2021).
- **SAKT:** SAKT is the first knowledge tracing model using the transformer structure, inspired by transformer in the field of NLP. The model's attention is used to identify different weights for time series to make more accurate predictions.
- **AKT:** AKT is a further developed version of the knowledge tracing model based on the attention mechanism, which significantly improves the prediction performance of the KT model and prevents the occurrence of over-parameterization utilizing the monotonic attention mechanism and Rasch encoding (Ghosh et al., 2020).
- **GKT:** GKT proposes a GNN-based knowledge tracing approach that reformulates the knowledge tracing task as a time-series node-level classification problem in GNNs (Nakagawa et al., 2019).

- **GIKT:** GIKT proposes a GNN-based knowledge tracing interaction model to address the limitations of data sparsity and multi-skill problems. More specifically, GIKT uses graph convolutional networks (GCN) to merge the relevance of problems and skills (Yang et al., 2020).
- **PEBG-DKT:** PEBG demonstrates that significant gains in KT can be achieved by pre-training embeddings for each problem on rich auxiliary features (Liu et al., 2020).
- **FBKT:** FBKT proposes a fuzzy Bayesian KT model to cope with continuous scoring scenarios to extend the applicability of the KT model (Liu et al., 2022).

Since the output of knowledge tracing models is usually binary values, this paper uses Area Under Curve (AUC) as the primary evaluation metric for all experiments of the KT task, and Root Mean Square Error (RMSE) and F1 are introduced as secondary metrics.

5.2. Experimental results

In this section, the HHSKT results are compared with the benchmark KT, and all experimental results are listed in Table 4. The model outperforms the other models on all three datasets, except for the Assist2017 dataset, where it obtains a slightly smaller AUC than the DKT model. In the Assist2009 dataset, the HHSKT model got the best average test AUC of 81.60%. We found that the model's prediction results were more accurate when the number of exercises per segment was lower. Experimental results for DKT and AKT showed that the attention mechanism accurately captured sequence dependencies. 78.30% and 81.5% were achieved for AKT and PEBG-DKT, respectively, indicating that the graph structure was more effective for multi-skill problems. The AUC of DAS3H is 3% higher than that of PFA, indicating that the forgetting effect of learners is a vital retrospective component of learners' ability. In Assistment2017, DKT and HHSKT achieved similar results. By contrast, the AUCs of DKT and GIKT are slightly lower than the HHSKT model at 83.30% and 83.80%, respectively. The results suggest that graph convolution is more suitable for this scenario. This may be due to a greater number of scaffolding issues in the Assistment2017 dataset, which limits the distance dependence of the original attention mechanism. In addition, we found that the performance of PFA was about 3% worse than IRT. The results suggest that sometimes when the learner interaction data is enormous (many learner interactions in Assistment2017 exceeded 1000), the volatility of correct and incorrect responses becomes insensitive, while information such as items can still provide accurate fit information. On Junyi15, HHSKT still achieved the best performance on the junyi15 dataset with an AUC of 82.60%. However, the difference with AKT was not significant, probably due to the relatively small number of knowledge points in the Junyi15 dataset. In particular, FBKT achieves similar results to PFA on this dataset, which indicates the effectiveness of FBKT. However, FBKT is not suitable for use as an online knowledge tracing method. In this task, graph neural network-based approaches outperformed regression-based models, such as GIKT and PEBG-DKT, by 10% on average in terms of AUC performance.

Table 4
Results.

Category	Methods	ASSIST2009		ASSIST2017		Junyi15	
		AUC	ACC	AUC	ACC	AUC	ACC
Regression-based model	IRT	0.672	0.679	0.680	0.664	0.721	0.831
	PFA	0.711	0.710	0.645	0.653	0.702	0.838
	FBKT	0.646	0.635	0.566	0.628	0.702	0.745
	DAS3H	0.742	0.723	0.716	0.690	0.767	0.843
Deep model	DKT	0.701	0.757	**0.833**	0.769	0.745	0.836
	SAINT	0.745	0.730	0.608	0.652	0.796	0.850
	SAINT+	0.749	0.732	–	–	–	–
	SAKT	0.739	0.706	0.706	0.685	0.793	0.850
	GKT	0.715	0.713	0.682	0.678	–	–
	GIKT	0.791	0.739	**0.838**	0.764	**0.825**	0.855
	PEBE-DKT	**0.815**	0.770	0.729	0.688	0.798	0.857
	AKT	0.783	0.737	0.747	0.698	0.810	**0.858**
	HHSKT	**0.816**	**0.749**	**0.842**	**0.757**	**0.826**	**0.852**

Table 5
Ablation results.

Methods	ASSIST2009		ASSIST2017	
	AUC	ACC	AUC	ACC
N-SerialAtt	0.804	0.741	0.811	0.738
N-RTE	0.803	0.741	0.835	0.767
N-ModifyAtt	0.788	0.735	0.771	0.719
N-Diff	0.809	0.744	0.824	0.752
N-GraphEmbed	0.799	0.739	0.816	0.748

5.3. Ablation results and comparison

5.3.1. Ablation study

In this section, a thorough ablation study is performed to demonstrate the validity of the key components of the HHSKT model. HHSKT was subjected to several ablation experiments to demonstrate how each part works and provide insight into the contribution of each part of the model to the whole. Five variants of the model were obtained by masking some modules in the model. All results were obtained by comparing the two datasets as shown in Table 5.

HHSKT consists of multiple modules. N-SerialAtt indicates that the serial position attention module is removed. N-RTE means that additional input information is removed, such as Resp Times or Lag Times; N-ModifyAtt means that all modified attention modules are removed, and only the attention of the original converter is retained; N-Diff means that the difficulty information estimates of the input items are not introduced; N-GraphEmbed means that all graph embedding features are removed. The ablation experiments of the above modules can further understand the intrinsic performance between different modules of the model. The ASSIST2009 and ASSIST2017 datasets have a different number of items and skills, so we conducted ablation experiments on the model on these two datasets (see Table 5).

N-ModifyAtt and N-SerialAtt: The experimental results from the ASSIST2009 dataset show that the AUC of the model decreases somewhat in both N-ModifyAtt and N-SerialAtt settings. Those results demonstrate that the serialized window attention mechanism is superior to the original attention mechanism and the monotonic attention mechanism. It also shows that learners' performance is more dependent on the performance of interactions within a short period in the KT task. The information of interaction sequences within a long time window interferes with the model's judgment of the learner's performance in the current moment.

N-RTE: The results of the N-RTE sub-model differed significantly from the original model AUC, suggesting that introducing additional information for each linked edge during the graph convolution process can better aggregate the target node information and thus improve the model capability.

N-Diff and N-GraphEmbed: The results from N-Diff and N-GraphEmbed demonstrate that the graph convolutional network learns more node representations by inputting natural graph structure features of interaction sequences, which is also consistent with our intuition that the same item or skill can be enhanced in its representation by aggregating different learner interaction features.

Fig. 5 shows the overall comparison results for different training ratios for this task. The results show that HHSKT outperforms the other baseline models. The results clearly show that the HHSKT model can take full advantage of the information contained in the different interaction sequences, which is beneficial to the prediction performance.

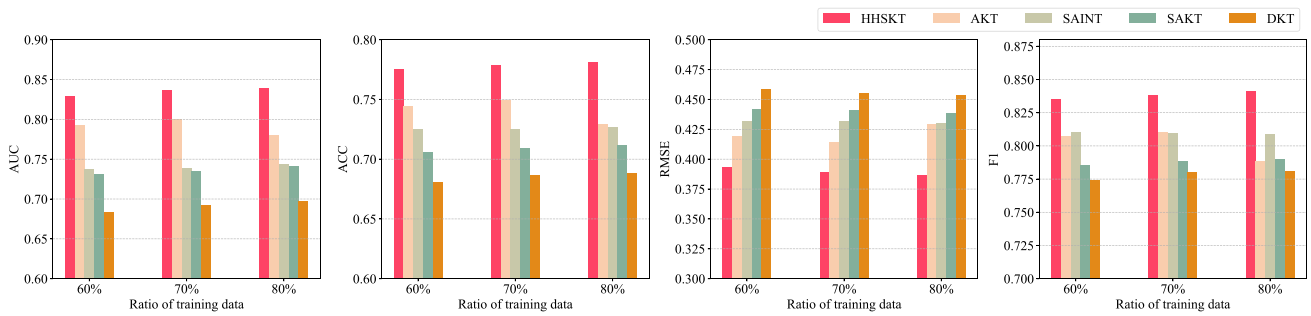


Fig. 5. The predicted results of learner performance for different training ratios under the four metrics.

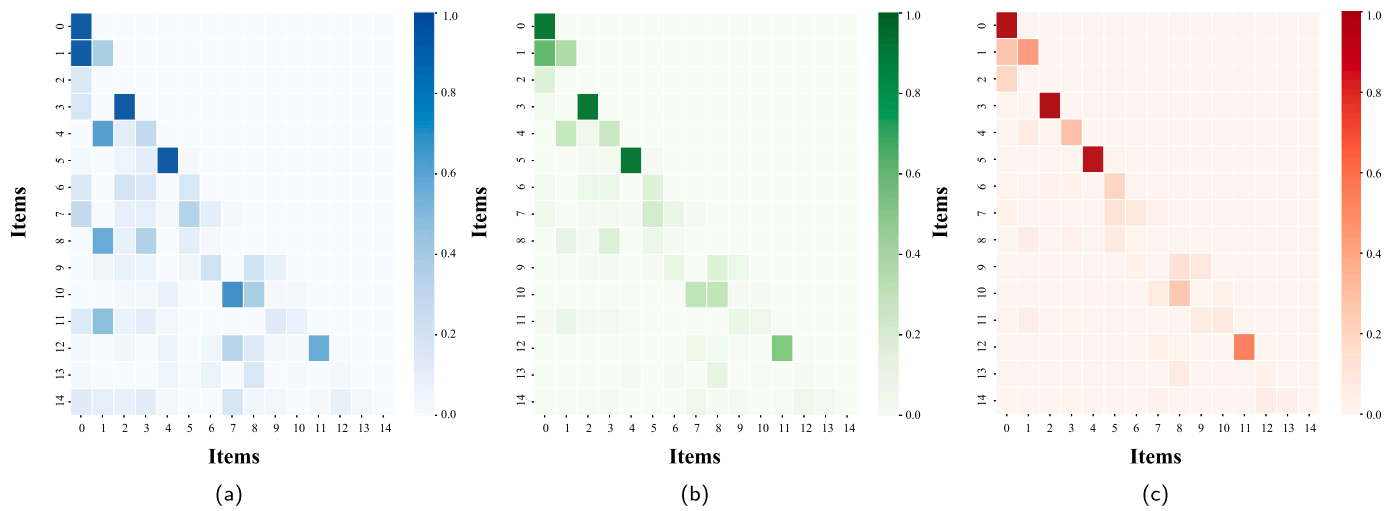


Fig. 6. (a) The original attention heatmap; (b) The attention heatmap comes from AKT; (c) Attention heatmap of HHSKT.

5.3.2. Analysis and visualization

In this section, the difficulty assessment structure and the base feature extraction structure are visualized to understand the model further. Batches of inputs from the training process of the ASSIST2009 dataset were selected, their overall processing flow was tracked, and their processing was visualized. The weighted heat maps of each of the three attention mechanisms are shown in Fig. 6. Only attention-weighted information for the first 15 interactions was selected for observation purposes.

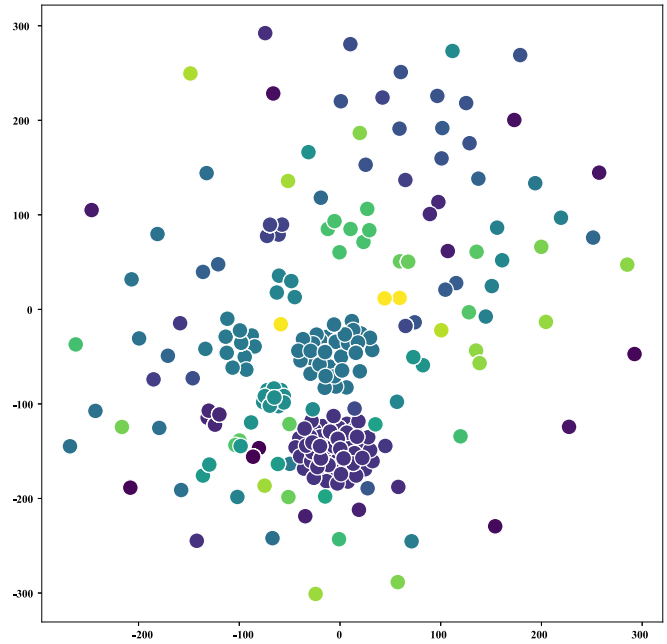
The original attention weights, AKT attention weights, and HHSKT attention weights are shown from left to right in Fig. 6. The monotonic attention mechanism is significantly shorter than the original attention weights, and the most recent interaction is given a higher weight than the other interactions. This can be attributed to the forgetting behavior of the learning process, where only the most recent interaction can inform the learner's knowledge state. Furthermore, comparing sequential and monotonic attention reveals that sequential attention places more emphasis on the relevance of the interaction between the current location of the current window and nearby locations and prevents out-of-window attention to information. It indicates that learners are more inclined to be influenced by learning interactions from the most recent period.

In addition, Graph embedding information has been shown to be effective in ablation experiments to improve the predictive power of the model. Items embedded in a batch of ASSIST2009 dataset were visualized using t-SNE, and items belonging to the same skill were labeled with the same color, as shown in Fig. 7. Different colors in Fig. 7(a) represent different aggregation categories, and compared with (b), it can be found that the abstract information represented by different aggregation regions has significantly different meanings, which confirms the effectiveness of our hierarchical feature extractor. The Fig. 7(b) shows the distribution relationship between different items after information aggregation, and it is obvious from the figure that multiple aggregation zones of different sizes are formed, which indicates that graph aggregation well distinguishes different items and assigns unique feature embedding information to them.

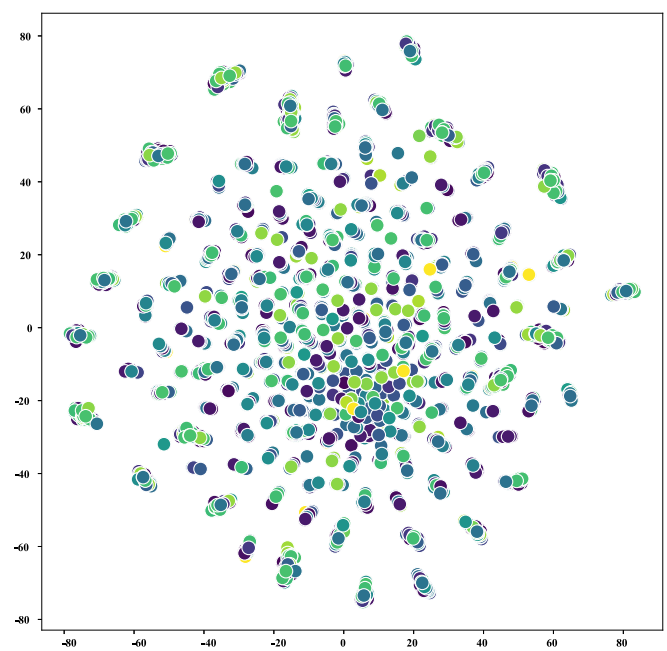
Fig. 8 shows the parameter scores given by the model after difficulty estimation for the ten questions belonging to the two concepts, respectively. The more significant overlap part in the figure represents the more similar items. The significant difference in the distributions of the two skills is evident in Fig. 8(b), which implies the need to differentiate the items' difficulty differences further when predicting the probability of mastery of items belonging to the two skills. The explanation derived in Fig. 8(a) is that there are still nuanced differences in difficulty for items attributed to the same skill. In addition, the item distribution also shows better consistency compared to the skill distribution.

6. Conclusion

In this study, we provide a relational representation of heterogeneous graphs for the KT problem in order to enhance the representation of input information and strengthen the contextual representation of input features. We additionally transform a matching-model-based difficulty estimate system in order to further discriminate between items by assessing the problem's difficulty. In addition, we simulate the forgetting behavior of learners using a positional decay kernel function and use an attention strategy based on sequential positional features. We proved the effectiveness of our suggested methodology by applying it to three real-world datasets. Experiments on three real-world datasets demonstrate that our model delivers state-of-the-art performance, with an average AUC gain of 1% to 3% over the baseline.



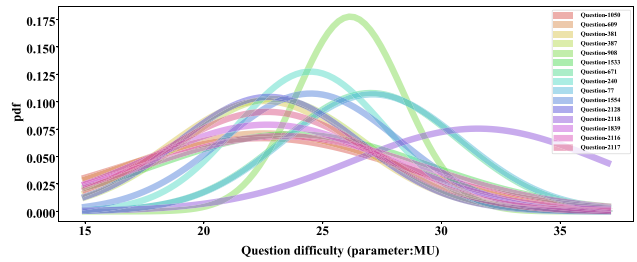
(a)



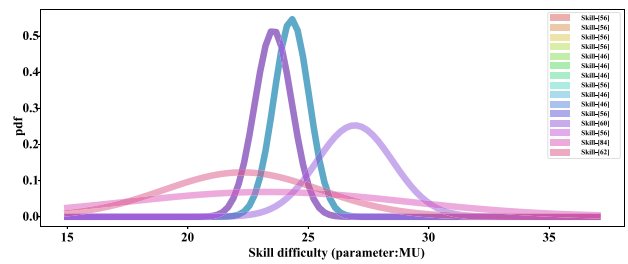
(b)

Fig. 7. Question relations and skill relations on the ASSISTments2009 dataset. (a) Skill embedding from t-SNE; (b) Question embeddings from t-SNE.

In the future, we intend to extract semantic and sentiment features from the question in order to improve the question representation. In addition, we intend to include information such as attempts and workouts in the graphical feature representation. Lastly, learner cognition while learning is progressive, and we also propose to trace learners' competency on different questions by tracing their cognitive changes from the perspective of the learners.



(a)



(b)

Fig. 8. The difficulty information uses a gaussian distribution to represent the beliefs about the learner. Each belief distribution can be uniquely represented using a mean μ (kurtosis) and a standard deviation σ (scatter). (a) Question difficulty parameters from Difficulty Estimation; (b) Skill difficulty parameters from Difficulty Estimation;.

CRediT authorship contribution statement

Qin Ni: Funding acquisition, Conception and design of study, Writing – review & editing. **Tingjiang Wei:** Acquisition of data, Methodology, Software, Writing – original draft. **Jiabao Zhao:** Supervision, Writing – review & editing. **Liang He:** Project administration, Resources. **Chanjin Zheng:** Writing – review & editing.

Declaration of competing interest

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Data availability

Data will be made available on request.

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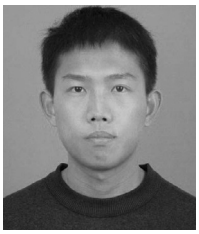

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